

Experiment No. 2

Aim

Socket Programming for Transport Layer Packets (TCP Server)

Theory

A **socket** is a software abstraction representing an endpoint of a two-way communication link between two programs running on a network. It is uniquely identified by the combination of an **IP Address** and a **Port Number**.

At the Transport Layer of the OSI model, TCP (Transmission Control Protocol) is used for connection-oriented, reliable, ordered, and error-checked data delivery. TCP Socket Programming on the server side operates as a **Passive Open** process, waiting for incoming connection requests from client nodes.

Server-Side Execution Steps:

1. **Socket Creation (socket()):** The server creates an endpoint for network communication using the domain AF_INET (IPv4) and socket type SOCK_STREAM (TCP).
2. **Binding (bind()):** The created socket is bound to a specific local IP address and port number using the bind() call. This assigns a designated location where clients can direct their connection requests.
3. **Listening (listen()):** The server places the socket into a passive listening state using listen(), maintaining a backlog queue for incoming client connection requests.
4. **Accepting Connection (accept()):** Upon receiving a request, the server executes accept(). This blocks execution until a client connects, completing the **TCP Three-Way Handshake** (SYN, SYN-ACK, ACK). Once accepted, a *new dedicated socket descriptor* is generated specifically for reading and writing data with that specific client.
5. **Data Transfer (recv() / send() or Streams):** Data exchange occurs reliably over stream sockets using input and output buffers.
6. **Closing Connection (close()):** The server terminates the connection using close(), executing the TCP four-wave termination procedure to release system resources.

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```
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO: ~
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO:~$ hostname -I
hostname-I: command not found
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO:~$ hostname -I
10.17.60.37 10.17.108.110
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO:~$ vim server1279.c
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO:~$ gcc server1279.c -o server
lab-3pc-27@lab-3pc-27-OptiPlex-3280-AIO:~$ ./server
TCP Echo Server listening on port 10279...
Received: hello
Received: hiii
Received: parth
```

```

#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <netinet/in.h>

#define PORT (10000 + 279) // Replace 'prn' with your PRN value
#define BUF_SIZE 1024

int main()
{
    int server_fd, new_socket;
    struct sockaddr_in address;
    char buffer[BUF_SIZE];
    int addrlen = sizeof(address);

    // Create TCP socket
    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    // Prepare server address
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);

    // Bind
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Listen
    listen(server_fd, 3);

    printf("TCP Echo Server listening on port %d...\n", PORT);

    // Accept client connection
    new_socket = accept(
        server_fd,
        (struct sockaddr *)&address,
        (socklen_t *)&addrlen
    );

    while (1)
    {
        memset(buffer, 0, BUF_SIZE);

        int bytes = read(new_socket, buffer, BUF_SIZE);

        if (bytes <= 0)
            break;

        printf("Received: %s", buffer);

        // Echo back to client
        send(new_socket, buffer, bytes, 0);
    }

    close(new_socket);
    close(server_fd);

    return 0;
}

```

Conclusion

The TCP Server socket was successfully implemented at the Transport Layer. The passive open execution sequence (socket, bind, listen, and accept) was executed, establishing a reliable, connection-oriented channel capable of processing incoming client data streams.

